

MAKERERE UNIVERSITY BUSINESS SCHOOL
FINAL EXAMINATION FOR THE
DEGREE OF BACHELOR OF BUSINESS COMPUTING OF
MAKERERE UNIVERSITY
ACADEMIC YEAR 2025/2026

COURSE NAME: OBJECT ORIENTED PROGRAMMING

COURSE CODE: BUC2125

YEAR OF STUDY: TWO

SEMESTER: ONE

DATE: 7th OCT, 2025

INSTRUCTIONS:

1. Submit your .java source files compressed in a single .zip folder named:
YourName_StudentOOPAssignment.zip
2. Include short comments explaining your code and each OOP principle demonstrated.
3. Ensure your program runs without compilation errors.
4. All work must be original and completed independently.
Use of code from classmates, online sources, or AI tools (e.g., ChatGPT, GitHub Copilot) is strictly prohibited.
Any violation will result in a zero grade and possible disciplinary action.
5. **Deadline for submission is Saturday 11th October at 00:00 AM.**

Assignment Overview

In this assignment, you are required to create a simple Java program that demonstrates the four main Object-Oriented Programming (OOP) principles — Encapsulation, Inheritance, Polymorphism, and Abstraction — while practicing the use of classes, objects, variables, constants, methods, and basic input/output using the command line.

You must structure your program cleanly and include short explanations where required.

Part A: Classes and Objects (10 Marks)

Task:

1. Create a class named **Person** in a package name of your choice with the following:

Instance variables:

- String name
- int age

A method `displayInfo()` that prints the name and age.

2. Create another class named **Student** that **inherits** from **Person**.

Add a variable:

- `double marks`

Add a **constant** named `SCHOOL_NAME` (e.g., "Makerere Business School").

Add a **constructor** to initialize name, age, and marks.

Override the `displayInfo()` method to include marks and the school name.

Part B: Methods, Variables, and Data Input (10 Marks)

Task:

In your main method:

1. Use the **Scanner** class to input the following from the command line:

Student's name

Age

Marks

2. Create a **Student** object using the input data.
3. Add a method **gradeStudent()** inside the **Student** class that returns:

"Excellent" if $\text{marks} \geq 80$

"Good" if $50 \leq \text{marks} < 80$

"Needs Improvement" if $\text{marks} < 50$

4. Display the student's details and grade on the console.

Part C: Demonstrating OOP Principles (20 Marks)

1. Encapsulation (5Marks)

- Make all the **Student** class variables **private**.
- Provide **getters and setters** to access and modify data safely.
- In a short explanation (as comments in your code), describe how encapsulation protects data from unauthorized modification.

2. Inheritance (5 Marks)

- Explain (in a short comment) how the Student class inherits properties and behaviors from the Person class, enabling **code reusability**.

3. Polymorphism (5 Marks)

- Demonstrate **method overriding** by redefining displayInfo() in the Student class.
- Include a short explanation of how **polymorphism** allows the same method to behave differently depending on the object type.

4. Abstraction (5 Marks)

- Create an **abstract class** named Learner with an **abstract method** study().
 - Let the Student class **extend** Learner and provide a concrete implementation of study() that prints a message like: "The student is studying for exams."

“ We Either Code Today or Debug Forever”